

The following questions are intended to guide you through the most important concepts that you must understand from the lectures and the material provided.

A good exercise would be to study the lectures/course material until you are able to answer them without having to read or search for it in the slides.

Lecture 2: Biometric systems, working modes, performance & errors

Questions: structure of biometric systems

- which are the main parts/modules of a biometric system (slide 4)
- give a brief description of the function of each part, commenting the most critical aspects to consider (slides 5-8)
- what is the difference between the enrolment and the authentication phases
- keywords of the above questions: features, template, score, threshold, enrolment...

Question: working modes of biometric systems (slides 11-14)

- which are the two possible working modes of a biometric system
- give a brief description of how each mode works, pointing out the steps involved in relation to the different parts of a biometric system
- keywords of the above questions: verification (1:1), identification (1:N, 1:N+1)

Question: computing performance of biometric systems (verification)

- Which are the two possible errors of a biometric system in verification mode (slide 19)
- Understand how these two errors can be computed (slides 20-23, and exercise of lecture 3)
- Understand the different graphs and concepts involved: histogram of genuine and impostors, threshold, FAR/FRR at a given threshold (how to measure), FAR/FRR curves, EER, DET curve
- Understand the implication of each error, to what users each error affects (slide 26)
- Understand the opposite behaviour of FAR and FRR, and which one we should minimize depending on the intended application (high security or high convenience)
- Understand the difference between similarity scores and distance scores (slide 27), and how this affects the different graphs and way of calculating errors

Question: computing performance of biometric systems (identification)

- Understand the possible errors involved in a system working in identification mode, both 1:N and 1:N+1 (slides 30-36)
- Understand the different graphs and concepts involved: TPIR, CMC curve
- Understand how these elements can be computed (exercise of lecture 3)
- Understand the impact of the value of N in identification mode (slide 37) and in which situations identification systems must be avoided (slides 38-39)

Question: other considerations regarding performance of biometric systems

- Understand the meaning of FTA and FTE errors (slide 41)